Visualisation has been done to examine the relationship between some of the predictor variables and the dependent variable. Bar plots have been used to visualize the mortality rate by admission type and also the mortality rate by first care unit in the train dataset. From the barplot, MICU care unit and emergency admission type has the highest mortality rate. Moreover, a box plot has also been used to see whether age affects mortality. As expected, higher age leads to higher mortality. In the first attempt where PCA is considered in the model, a scree plot has also been used to see how many PCA should be included. As a result, the elbow point is around 5-10 and in the model, the first 5 PCA has been included at first.

Before testing the model, data was being reprocessed to ensure the accuracy of the prediction. At first, the secondary causes of patients are not included in the main dataset. After merging train_x and train_y data, the age of which patients entered the ICU was calculated through the variable ADMITTIME and DOB. For patients over 89 years old, exact birth dates are considered personally identifiable information (PII) and so DOB and admittime has been readjust. Due to this, if the entered_age value is larger than 89, I purposely set it to 90. After all of this has been done, irrelevant information such as subject_ID and more has been removed. Moreover, categorical variables such as gender and admission type were also converted to factors.

However, after a few trials, the roc AUC of the out sample prediction are below expectations and so secondary causes are being included in the test and train set for higher accuracy. Firstly, in order to incorporate the secondary causes into the model, the ICD9_CODE data has been pivoted to a wider format. This involved transforming the dataset so that each unique ICD-9 code became a separate binary column, indicating whether or not a patient was diagnosed with that condition. This approach allows the model to utilize the presence of specific diagnoses as features, thereby capturing more clinical information that may contribute to improved prediction accuracy. The same also applies to the ICD9_diagnosis column in the main dataset.

A simple logistic model has been used at first to predict the outcome. The current model, which uses secondary causes as predictors, achieved an out-of-sample ROC-AUC of 0.83 and in sample of 0.86. This shows an improvement over a previous logistic regression model that relied on principal component analysis (PCA) for dimensionality reduction, which yielded an AUC of 0.81 and in sample AUC of 0.78. The improvement suggests that directly incorporating secondary causes preserves important clinical information that PCA may miss, leading to better model performance. The in-sample AUC (0.78) being slightly lower than the out-of-sample AUC (0.81) is unusual but despite this, the out-of-sample AUC reliably reflects the model's generalization performance.

At first, a simple decision tree model was used for prediction. However, due to persistent computational limitations and instability encountered during the hyperparameter tuning process, the decision tree model could not be successfully trained on the full dataset. Given the complexity and high dimensionality of the data, a single decision tree may also be insufficient for capturing non-linear relationships and interactions. Therefore, more advanced ensemble methods such as random forests and boosted trees were explored as more suitable alternatives for predictive modeling.

The random forest model demonstrated near-perfect performance on the training data (AUC of 0.9998), but its test performance dropped to 0.937, indicating overfitting. In contrast, the boosted tree achieved more balanced results, with a training AUC of 0.9514 and a test AUC of 0.902, reflecting better generalization. While the random forest shows signs of overfitting, its test performance (AUC 0.937) still outperforms the boosted tree (test AUC of 0.902). This suggests the random forest remains the stronger model overall, despite learning the training data too perfectly.

For neural network models, different types of input data, including PCA-transformed ICD-9 codes and raw clinical features such as secondary causes have been used. The models consisted of two hidden layers with 64 and 32 nodes, ReLU activation functions, dropout layers to reduce overfitting, and a sigmoid output layer for binary classification. Various settings such as learning rate, batch size, and number of epochs have been tuned to improve performance. This architecture was chosen to balance model complexity with training stability, especially considering the high dimensionality of the clinical data. When trained on PCA-transformed features, the neural network performed well, achieving an in-sample AUC of approximately 0.77 and an out-of-sample AUC of 0.814. However, when using raw secondary cause features, the performance dropped significantly. Despite tuning the network structure and hyperparameters, the model trained on these raw features only reached an in-sample AUC of 0.5 and out-of-sample AUC between 0.429 and 0.623.

AI Reference

In this project, AI tools have assisted me in gaining a deeper understanding of neural networks, including the importance of incorporating techniques such as early stopping to improve prediction performance. I have also sought advice on determining the appropriate number of nodes and configuring the output layer for the neural network.